COMPUTER ENGINEERING · UNIVERSITY OF GEORGIA

I like building things.

I'm a computer engineering student and undergraduate researcher at the University of Georgia. I build embedded hardware, robotics, and systems software.

CURRENT FOCUS

- SERS lab research
- AI model evaluation
- Embedded systems
- Systems programming
- Robotics & autonomy

Selected Projects

Embedded hardware, backend systems, robotics, and computer vision.

EMBEDDED SYSTEMS · MOTOR CONTROL 01

Brushless Motor Controller

C · STM32 · KiCad · SPI

Designed a three phase motor controller PCB in KiCad and wrote C firmware for PWM, SVPWM, and field oriented control. The board has USB-C connectivity and supports an optional high resolution SPI absolute encoder.

Verified PWM timing and deadtime with an oscilloscope, then debugged gate drive and motor control behavior on the assembled board.

Firmware · PCB Design · FOC · SVPWM · Hardware Debugging

View project →

SYSTEMS SOFTWARE · SECURITY 02

Soltix.cc

Founder & Developer · Jan 2022–Jul 2025

Founded a commercial cheat detection service that analyzed Windows process memory and system artifacts. It reached more than 300 users and 37 customers.

Built a Windows driver in C/C++ and process analysis tools in Go, and administered Linux systems for customer deployments.

Windows Driver · Process Memory · C/C++ · Go · Linux

View project →

ROBOTICS · AUTONOMY 03

FRC Team 1683

Team Member · Aug 2021–May 2025 · Programming Lead for 2 years

Led programming for two of my four years on the team, developing autonomous and computer vision software for competition robots. The team of 85 students reached the FIRST World Championship in both years I served as lead.

Also contributed to robot design, electrical systems, fabrication, and CNC machining.

Robot Software · Autonomous Systems · Computer Vision · Electrical Systems · CNC

View project →

BACKEND DEVELOPMENT · HACKGT 13 04

SideQuests

MongoDB Prize · 1st Place, NSA HEARSAY

Built the Go backend and server infrastructure for a MongoDB backed app that combines time and location aware activity plans with shared outings, interest matching, and group chat.

Go | MongoDB | Backend Development | Server Infrastructure

View project → **Devpost ↗**

COMPUTER VISION · MECHANISM DESIGN 05

Sortify

Google Gemini Track Winner · MakeMIT x Harvard

Developed computer vision software to classify waste and pass the results to sorting controls. Designed a compact differential mechanism to route each item to the correct bin.

Computer Vision | Sorting Controls | Mechanism Design

View project → **Devpost ↗**

BACKEND DEVELOPMENT · DOCUMENT SEARCH 06

DocDoctor

MongoDB Track Winner · Google Cloud Honorable Mention · AI ATL

Built a Go backend and WebSocket pipeline for live support call transcripts and agent annotations. Added MongoDB persistence and document search APIs so agents could retrieve relevant material during calls.

Go | WebSockets | MongoDB | Document Search | APIs

View project → **Devpost ↗**

AI · EDUCATION 07

Sophi

NexHacks · Carnegie Mellon

Built an AI learning tool that generates questions in an instructor's style from course material and gives targeted hints to help students practice problem solving with material tailored to their class.

AI | Learning Tools | Education

View project → **Devpost ↗**

ROBOTICS · COMPUTER VISION 08

Distributed Planetary Analysis Platform

3rd Place · Autonomous Track · RoboTech

Developed software for a humanoid robot and autonomous vehicle using inverse kinematics, fiducial detection, pathfinding, PWM motor/servo control, and obstacle avoidance. The autonomous vehicle ran on a Jetson and used fiducial tracking to navigate its environment.

Jetson | Computer Vision | Inverse Kinematics | Pathfinding

View project → **Devpost ↗**

RESEARCH & WORK

Experience

B.S. in Computer Engineering at the University of Georgia.

01	**Mercor Intelligence**	**Independent Contractor · Jun 2026–Present.** Evaluated and ranked AI model outputs across prompts and tasks, validated correctness and instruction following, and created edge case prompts to test model limitations and identify areas for improvement.
02	**UGA SERS Lab**	**Undergraduate Researcher · Sep 2026–Present.** Designed the scanner architecture and modeled the complete mechanical system in Onshape for an automated Raman/SERS platform. Selected and ordered electronic components, corrected prior wiring faults, and contributed to fabrication and full stack scanner software.
03	**Soltix.cc**	**Founder & Developer · Jan 2022–Jul 2025.** Built commercial Windows process analysis software, including a C/C++ driver and Go tools, and managed Linux deployments for 37 customers.
04	**FRC Team 1683**	**Team Member · Aug 2021–May 2025 · Programming Lead for 2 years.** Led autonomous and computer vision development on a team of 85 students that reached the FIRST World Championship in both years I served as lead.

Technical Skills

Tools and disciplines used in my research, projects, and engineering work.

Languages

C, C++, Go, Python, Java, JavaScript

Engineering

Embedded systems, firmware development, PCB design, electronics, robotics, CAD/CAM, Onshape

Software & systems

Systems programming, Windows driver development, computer vision, backend development, WebSockets, MongoDB, Linux

Maxim Iliev
University of Georgia · B.S. Computer Engineering
maxim.iliev.a@gmail.com · github.com/MqxS · LinkedIn